

Venkata Saratchandra Patnaik Markonda Patnaikuni

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Cloud-native software implementation engineer with production experience in Kubernetes (EKS), Docker, GitOps, observability, hybrid/edge systems (multi-cloud/on-prem), Go/Python, GitOps (ArgoCD), Linux, and reliability-focused deployments and troubleshooting.

EDUCATION

Master of Science in Computer Science (CGPA: 3.81/4)

December 2025

Arizona State University, Tempe, AZ

Bachelor of Technology in Computer Science (CGPA: 9.11/10)

May 2022

Christ University School of Engineering, Bengaluru, India

PROFESSIONAL EXPERIENCE

Software Implementation Engineer: Amagi Media Labs, Bengaluru, India

August 2022 - November 2023

- Operated and improved a Kubernetes-based microservices platform (15+ services) on AWS EKS using Docker and Linux, increasing deployment frequency by ~30% through stronger release workflows and reliability practices.
- Implemented GitOps CI/CD with ArgoCD and standardized rollout/rollback procedures, reducing deployment risk and accelerating recovery during regressions.
- Automated customer/creator onboarding and integrations using Terraform and Python-based REST API workflows, cutting manual setup by ~90% and reducing CDN delivery failures by ~95%.
- On-call/DRI (production): resolved a critical client streaming incident where audio loss triggered fallback content by tracing concurrent inbound streams and implementing automatic failover to a healthy stream after a defined audio-silence window (less than 1 hour to resolution).
- Drove incident triage and platform troubleshooting using Grafana + Loki for observability; diagnosed Kubernetes pod/service health issues and applied configuration fixes/maintenance restarts to restore operator dashboard functionality.

Web Development Intern: Blueplanet Solutions Inc., India April 2021 - June 2021

- Optimized MySQL database using stored procedures for 'Campus Club', improving query speed by **50%** and enabling faster data access for 1,000+ student profiles.
- Enhanced a responsive student search UI with HTML, CSS, JavaScript, and PHP, reducing search time by **60%** and improving user experience.
- Analyzed web server logs to identify memory leaks, enhancing application stability by 30% and reducing server resource usage.

PROJECTS

Real-Time Face Recognition on AWS Greengrass (Cloud Computing Project)

- Built an edge-to-cloud pipeline using AWS IoT Greengrass + MQTT + Lambda/SQS, simulating smart camera to core communication over EC2; debugged distributed components via CLI/logs and automated deployments with shell scripts.
- Deployed MTCNN and FaceNet models under pip-free constraints by integrating raw PyTorch source, optimizing runtime within AWS Greengrass component limits.
- Debugged a complex distributed pipeline using CLI tools and logs across Greengrass, MQTT, and Lambda; automated deployments via shell scripts and passed over 100+ real-time autograder test cases.

TECHNICAL SKILLS

Programming & Scripting Languages: Python, R, Java, Kotlin, Go, JavaScript, TypeScript, C

Cloud & DevOps Tools: AWS(S3, EKS, EC2, Greengrass, Lambda, Load Balancer), Docker, Kubernetes, Linux, Shell Scripting

CI/CD, Monitoring & Dev Tools: GitHub, GitOps, ArgoCD, Jira, Jenkins, Grafana

Web & Mobile Development: React.js, Node.js, Express.js, HTML/CSS, Bootstrap, Android Studio, Flask, REST APIs

Databases: MySQL, PostgreSQL